HGVRec: Action-Segment Hypergraph Aggregation with Role-aware Entity–Action Fusion for Video Recommendation

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Abstract

Multimodal large language models (MLLMs) provide high-level video semantics that can complement metadata and interaction signals in recommendation. Privacy concerns around server-side analysis of viewing content motivate on-device processing with lightweight MLLMs. The challenge is to organize the extracted information into representations useful for recommendation. We introduce HGVRec, a video representation learning framework that uses explicit structure to improve the recommendation utility of MLLM-generated scene graphs. Action-Segment Hypergraph Aggregation (ASHA) models scenes as nodes and maximal contiguous runs sharing a normalized action key as hyperedges; it aggregates information from scenes to segments and back, and incorporates segment extent into video readout. Role-aware Entity–Action Fusion (REAF) enriches action features with entity features according to extracted actor, target, tool, and place roles, providing the scene representations aggregated by ASHA. The representation modules are trained jointly with a sequential recommender using frozen text embeddings of the extracted scene information. On MicroLens-100K and M^3L-10M, HGVRec improves NDCG@10 over a title-only encoder by 28.90% and 3.38%, and over scene-description embeddings by 14.51% and 2.68%.

CCS Concepts

• **Information systems** → **Recommender systems**; • **Computing methodologies** → *Learning latent representations.*

Keywords

Video Recommendation, Hypergraph Learning, Scene Graph, Multimodal Large Language Models

1 Introduction

Video recommendation on Web platforms depends on representations that connect visual content with user preferences. Multimodal large language models (MLLMs) provide semantic information beyond item metadata: DWYS [2] uses generated descriptions as recommendation inputs, while IISAN [4] adapts multimodal representations through side networks. Privacy concerns around server-side analysis of viewing content favor on-device extraction with compact MLLMs. This raises a representation learning question: how can a recommender make better use of the information that a compact MLLM already extracts?

We investigate explicit structure as a way to improve the recommendation utility of compact MLLM outputs. A scene may contain entities and actions, yet their separate aggregation leaves participation roles implicit; likewise, aggregating scene vectors without action-level organization leaves continuity implicit. For example, a person may cut a carrot with a knife across several consecutive scenes (Figure 1). The person, carrot, and knife play different roles in the same action, while the repeated action links the scenes into a coherent segment.

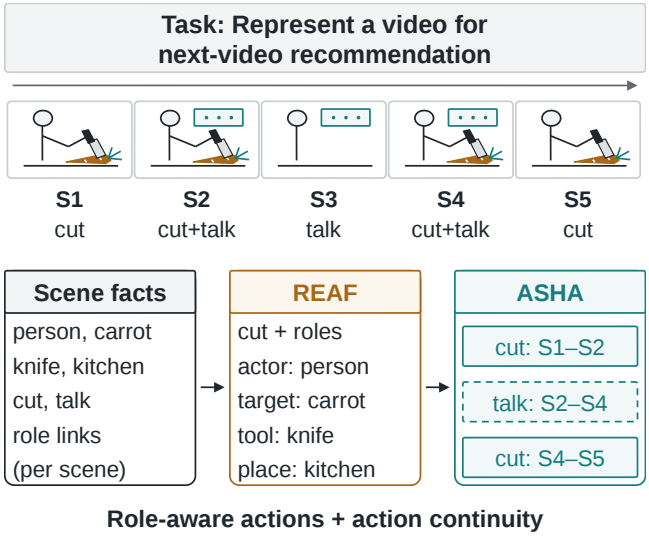


Figure 1: An illustrative example. REAF encodes who acts on what and with which tool; ASHA groups consecutive scenes sharing an action, so the two cut segments stay separate.

We introduce HGVRec, a hierarchical framework centered on Action-Segment Hypergraph Aggregation (ASHA). Building on the general aggregation framework of AllSet [1], ASHA treats scenes as nodes and maximal contiguous runs sharing a normalized action key as hyperedges; scenes with multiple actions can belong to overlapping segments. Messages pass from scenes to segments and back, with segment extent incorporated into the representations and video readout. Role-aware Entity–Action Fusion (REAF) supplies ASHA with scene features by fusing entity features into actions according to extracted actor, target, tool, and place assignments. Our contributions are: (1) a framework that explicitly organizes the semantics extracted by a compact MLLM into role-aware scenes and action segments for recommendation; (2) contiguous action segments as an aggregation unit for video representation; and (3) experiments on MicroLens-100K and M^3L-10M in which HGVRec outperforms title-only, scene-description, and mean-pooled scene-graph encoders, with ASHA accounting for most of the gain.

2 Related Works

Video recommendation benefits from modality-specific graph representations, as in MMGCN [12], MLLM-generated descriptions, as in DWYS [2], and representations from frozen MLLMs fused with

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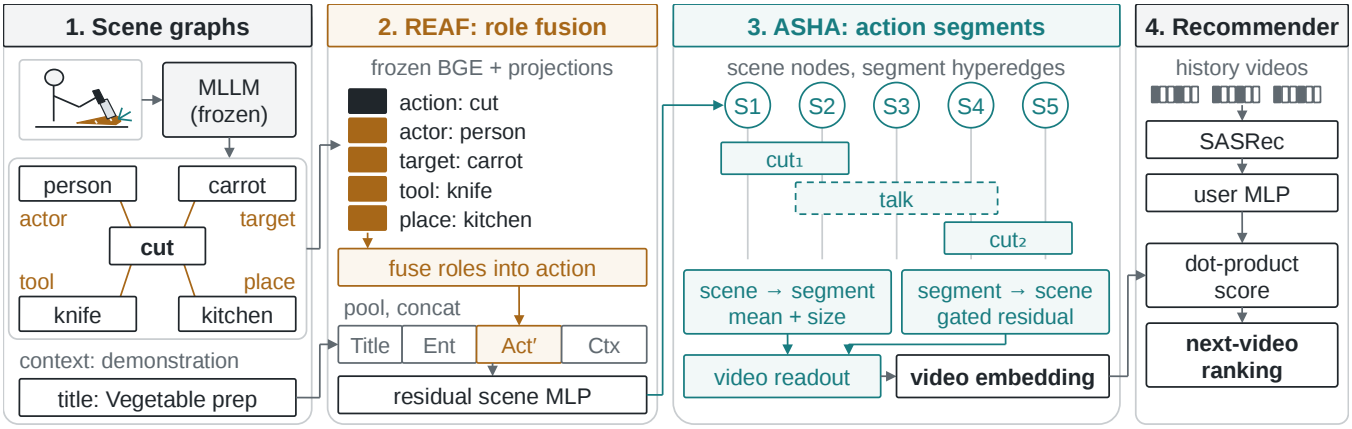


Figure 2: Overview of HGVRec on the cooking example. (1) A frozen MLLM turns each scene into a graph whose action links entities as actor, target, tool and place. (2) REAF fuses the role-linked entity features into the action before pooling a scene vector. (3) ASHA groups scenes into contiguous action segments (hyperedges), passes messages between scenes and segments, and reads out the video. (4) The same encoder embeds history and candidate videos for a SASRec recommender.

item IDs, as in DFF [11]. Hypergraphs also support recommendation: DHCN [13] connects items through sessions, and MHCR [7] combines multiple graph views with hypergraph contrastive learning; HGVRec instead constructs a hypergraph within each video, with scenes as nodes and contiguous action segments as hyperedges. ASHA follows the node–hyperedge aggregation pattern of HGNN [3] and AllSet [1], using action continuity to define overlapping groups of scenes and incorporating segment extent. Structured visual representations model entities in semantic roles [9] and actions through spatiotemporal scene graphs [5]; REAF uses extracted actor, target, tool, and place assignments to fuse entity features into action representations for recommendation.

3 Methods

3.1 Problem Formulation and Overview

Let $\mathcal{V}$ denote the candidate video catalog and $\mathcal{H}_u$ the observed interaction history of user u. HGVRec learns a content representation $\mathbf{z}_v \in \mathbb{R}^D$ for each video and a user representation $\mathbf{q}_u \in \mathbb{R}^D$ from the history, and ranks candidates for the next interaction by $f(u, v) = \mathbf{q}_u^\top \mathbf{z}_v$; the same video encoder supplies history-item and candidate-item representations. We observe each video through temporally ordered windows of sampled frames and call a window a *scene*; this does not imply a detected shot or a ground-truth action boundary. A frozen MLLM produces a scene graph containing entities (name and attributes), action phrases with actor, target, tool, and place assignments, and a scene context (format, medium, and topics). After structural validation, video v has S valid scenes with original window indices $p_1 < \cdots < p_S$. The video title enters the representation stage as a separate input after visual extraction. We introduce ASHA first, taking scene vectors $\mathbf{x}_s \in \mathbb{R}^D$ as inputs, and then define their construction through REAF (Figure 2). Throughout, $[\cdot\,;\cdot]$ denotes concatenation and σ denotes ReLU.

3.2 Action-Segment Hypergraph Aggregation (ASHA)

A deterministic function κ converts an action phrase into a key: it lowercases and tokenizes the phrase, removes participant markers and a fixed set of function words, and normalizes the first remaining token using irregular-form mappings and suffix rules. This is a lightweight lexical rule that does not merge synonymous actions. For key k, an *action segment* is a maximal run of scenes that contain k and have consecutive original indices ($p_{s+1} = p_s + 1$). A missing window, a scene without k, or a later reappearance of k separates segments; singleton segments are retained, and a scene containing several keys belongs to several segments. A segment thus expresses continuity of an extracted action key rather than a verified continuous real-world event. The hypergraph has one node per valid scene and the K segments as hyperedges, with fixed incidence $H_{sk} = \mathbb{I}[s \in e_k]$, segment sizes $n_k = \sum_s H_{sk}$, and scene degrees $d_s = \sum_k H_{sk}$.

Following the two-stage aggregation framework of AllSet [1], ASHA pools each segment and augments it with its extent, updates each scene with the mean of its incident segments, and reads out the video:

$$\mathbf{u}_k = \sigma\Big(W_{\text{seg}}\big[\tfrac{1}{n_k}\textstyle\sum_s H_{sk}\mathbf{x}_s;\ \log n_k;\ n_k/S\big] + \mathbf{b}_{\text{seg}}\Big), \tag{1}$$

$$\mathbf{x}'_s = \mathbf{x}_s + \alpha\,\mathbb{I}[d_s > 0]\,\sigma\Big(W_{\text{msg}}\big[\mathbf{x}_s;\ \tfrac{1}{d_s}\textstyle\sum_k H_{sk}\mathbf{u}_k\big] + \mathbf{b}_{\text{msg}}\Big), \tag{2}$$

$$\mathbf{z}_v = \tfrac{1}{S}\textstyle\sum_s \mathbf{x}'_s + \beta\,\mathbb{I}[K > 0]\,W_{\text{out}}\tfrac{1}{K}\textstyle\sum_k \mathbf{u}_k. \tag{3}$$

The extent features give the number of member scenes and its fraction of the video's valid scenes. The scalars α and β are trainable and initialized to zero, so ASHA starts as a mean of its input scene vectors; videos without segments keep the scene mean.

3.3 Role-aware Entity–Action Fusion (REAF)

Let $\phi(\cdot)$ be a frozen text encoder. We independently project the video title, each entity description, each action phrase, and the

Table 1: Recommendation performance (mean±sample standard deviation over seeds 42, 43, and 44). ΔN and ΔH are relative changes (%) in NDCG@10 and HR@10 over Title-only. Params counts trainable parameters including the sequential recommender. Bold marks the highest mean.

		MicroLens-100K				M^3L-10M			
Method	Params	NDCG@10	HR@10	ΔN	ΔH	NDCG@10	HR@10	ΔN	ΔH
Title-only	7.62M	.0337±.0005	.0642±.0009	–	–	.1388±.0019	.2324±.0024	–	–
Scene Description	7.62M	.0379±.0012	.0726±.0016	+12.56	+13.12	.1397±.0017	.2340±.0018	+0.68	+0.69
Scene Graph	7.62M	.0351±.0011	.0687±.0019	+4.25	+6.97	.1381±.0007	.2324±.0007	−0.47	−0.03
Scene Graph + ASHA	8.68M	.0422±.0010	.0805±.0019	+25.25	+25.41	.1423±.0017	.2398±.0019	+2.50	+3.16
HGVRec	8.76M	**.0434**±.0004	**.0822**±.0010	+28.90	+28.07	**.1435**±.0006	**.2405**±.0006	+3.38	+3.45

scene context into d dimensions with separate learned linear maps of ϕ, obtaining $\mathbf{t}_v$, $\mathbf{e}$, $\mathbf{a}$, and $\mathbf{c}_s$. For each action a and role r (actor, target, tool, or place), the role vector $\mathbf{g}_{a,r}$ is the feature of the linked entity, or a learned state $\mathbf{b}_r^{\text{abs}}$ or $\mathbf{b}_r^{\text{unk}}$ when the role is absent or unknown. With $\mathbf{r}_a$ the concatenation of the four role vectors in this fixed order, REAF updates the action residually:

$$\tilde{\mathbf{a}} = \mathbf{a} + \sigma(W_R[\mathbf{a}; \mathbf{r}_a] + \mathbf{b}_R). \tag{4}$$

The fixed slot order permits different transformations of the same entity feature when it serves different roles; entity features themselves are unchanged. Let $\bar{\mathbf{e}}_s$ be the mean of projected entity features and $\bar{\mathbf{a}}_s$ the mean of updated action features in scene s (zero if empty). The scene representation, with $D = 4d$, is

$$\mathbf{x}_s = \mathbf{b}_s + \sigma(W_{\text{scene}}\mathbf{b}_s + \mathbf{b}_{\text{scene}}), \quad \mathbf{b}_s = [\mathbf{t}_v; \bar{\mathbf{e}}_s; \bar{\mathbf{a}}_s; \mathbf{c}_s]. \tag{5}$$

A video with no valid scenes applies the same scene MLP to $[\mathbf{t}_v; \mathbf{0}; \mathbf{0}; \mathbf{0}]$, so extraction failures do not remove items from the catalog.

3.4 Sequential Recommendation and Joint Optimization

For each prediction event, we retain the most recent L history items, add learned position embeddings to their video vectors, and apply a causally masked SASRec-style Transformer [6]. The hidden state at the last valid position is passed through a two-layer residual user MLP to obtain $\mathbf{q}_u$; the content encoder has no item-ID embedding table. We train with an in-batch softmax cross-entropy over history–target pairs $(\mathcal{H}_i, y_i)$ using popularity-corrected logits $\mathbf{q}_i^\top \mathbf{z}_{y_j} - \log Q(y_j)$, where Q is the normalized target frequency among training transitions. Negatives that equal the positive or occur in the input history are masked, and evaluation ranks the uncorrected scores. All projections, role-state embeddings, REAF, the scene MLP, ASHA, the sequential encoder, and the user MLP are optimized jointly, while the MLLM, the text encoder, and the segment memberships stay fixed.

4 Experiments

We evaluate HGVRec through three questions: **RQ1** how it compares with title-only, scene-description, and mean-pooled scene-graph representations; **RQ2** what changes when ASHA is added to the scene-graph encoder; and **RQ3** what changes when REAF is added to the encoder that already uses ASHA.

Table 2: NDCG@10 differences ($\times 10^{-3}$), relative change (%) in parentheses, and Bonferroni-corrected 99.5% paired-bootstrap intervals.

Contrast	MicroLens-100K	M^3L-10M
HGVRec − Title-only	9.74 (28.90) [8.87, 10.59]	4.69 (3.38) [3.38, 5.95]
HGVRec − Scene Desc.	5.51 (14.51) [4.71, 6.30]	3.75 (2.68) [2.46, 5.01]
+ASHA − Scene Graph	7.08 (20.14) [6.34, 7.80]	4.13 (2.99) [2.91, 5.34]
HGVRec − +ASHA	1.23 (2.91) [0.53, 1.92]	1.22 (0.85) [0.03, 2.42]

4.1 Experimental Setup

Datasets and protocol. We use MicroLens-100K for its micro-video interactions and video content [8], and M^3L-10M, which adds multimodal movie information, including trailers, to MovieLens-10M [10]. For M^3L-10M, every rating is treated as an interaction, movies without usable trailer videos are excluded, and the task is predicting the next movie in a rating-event sequence. Users must have at least five interactions, and we retain their 13 most recent events; the candidate catalog is the union of items in these retained sequences. This yields 100,000 users, 19,671 items, and 678,355 interactions for MicroLens-100K, and 69,878 users, 8,197 items, and 908,412 interactions for M^3L-10M. The last event of each user is held out for test and the penultimate event for validation. Validation selects the checkpoint with the highest HR@10; test uses it without refitting, ranking the full catalog after masking all items in the preceding history.

Representation variants. All variants use the same content-based sequential recommender. *Title-only* projects the title embedding and applies a residual item MLP. *Scene Description* combines a projected title with projected scene-description embeddings, applies a residual scene MLP, and averages the scenes. *Scene Graph* projects title, entity, action, and context features, pools entities and actions separately within each scene, and averages the encoded scenes. *Scene Graph + ASHA* adds the action-segment module, and *HGVRec* further adds REAF. The three graph variants share identical extracted graphs, which are valid for 96.04% and 95.92% of the scenes in the two datasets; descriptions are a separate extraction. Scene graphs and descriptions are extracted with Qwen3-VL and embedded with a frozen BGE text encoder. We train with AdamW (weight decay 0.1, gradient-norm clipping 5.0) and seeds 42, 43, and 44.

Metrics and statistics. We report the mean and sample standard deviation of NDCG@10 and HR@10 over the three seeds, with

relative changes over Title-only. For NDCG@10, we average each user's metric over seeds and perform a paired bootstrap over users with 10,000 replicates; Bonferroni correction over all ten pairwise comparisons per dataset at family-wise level 0.05 yields 99.5% confidence intervals. HR@10 is reported descriptively.

4.2 Results

Overall performance (RQ1). HGVRec achieves the highest mean NDCG@10 and HR@10 among the five variants on both datasets (Table 1), with larger relative gains on MicroLens-100K. Its NDCG@10 intervals against Title-only and Scene Description exclude zero on both datasets (Table 2). Scene Graph alone is below Scene Description on both datasets, indicating that producing graph-formatted outputs and pooling their components is insufficient by itself to reproduce the gains of HGVRec.

Effect of ASHA (RQ2). With identical graph inputs, adding ASHA to Scene Graph raises NDCG@10 by 20.14% and 2.99%; both corrected confidence intervals are positive, and mean HR@10 also increases. The comparison adds the complete module (action-segment membership, size-aware segment encoding, scene updates, and segment readout) and its 1.05M trainable parameters.

Effect of REAF (RQ3). Adding REAF to Scene Graph + ASHA increases NDCG@10 by a further 2.91% and 0.85% with 83K additional parameters, and both corrected intervals exclude zero. This supports the incremental benefit of encoding actor, target, tool, and place assignments before pooling actions in the presence of ASHA.

5 Conclusion

We presented HGVRec, a video representation learning framework that organizes MLLM-extracted semantics through explicit structure: ASHA groups scenes into maximal contiguous action segments and aggregates them through a hypergraph, while REAF encodes entity participation roles within each scene. On MicroLens-100K and M^3L-10M, HGVRec achieves the highest mean NDCG@10 and HR@10 among the five evaluated variants; with shared graph inputs, adding ASHA improves NDCG@10 on both datasets, and adding REAF provides further gains when ASHA is present.

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